ENBC332: Homework #4

Due: Monday, 10/06/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 30

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: “problem1_fullname.R”. Start with the problem number and followed by your full name, e.g., “**problem1_jimmyazarnoosh**”.

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Note: All functions are recommended to follow the format below:
e.g., calculating range.

```
# ---------------------------------------------------
Data <- trees$Height
my_range_func <- function(Data){
  Range <- max(Data) - min(Data)
  return(Range)
}
# ---------------------------------------------------
```

Describing Relationships

Problem 1. Using the dataset from Homework 2, complete the tasks below. Your dataset should consist of two columns: the first column representing the x-axis and the second column representing the y-axis. (30 points)

a) Calculate **covariance** for your dataset using R function "`cov()`" and also write your own custom functions. (10 points)

1) Write a function to calculate covariance using a **for loop**.
2) Write a function to calculate covariance using the **sum()** function.

Note: You should write two separate functions to calculate covariance.

b) Calculate **Pearson's correlation coefficient** for your dataset using R function "`cor()`" and also write your own custom functions. (10 points)

1) Write a function to calculate Pearson's correlation coefficient using a **for loop**.
2) Write a function to calculate Pearson's correlation coefficient using the **sum()** function.

Note: You should write two separate functions to calculate correlation coefficient.

Note: The formula for the correlation coefficient is defined based on covariance and standard deviation. Use the custom covariance function you wrote in part (a), and the standard deviation function from Homework 2. Be consistent with the functions you use. If you write a function to calculate the correlation coefficient using a for loop, ensure you also use the for loop versions of the covariance and standard deviation functions.

c) Plot a **scatter plot** of the first column (x) versus second column (y) to visualize their relationship. Using **ggplot** is optional, but feel free to use it if you prefer. Please answer the following questions: (10 points)

1) What does the sign of the covariance indicate about the relationship between the two variables?
2) What does the magnitude of the correlation coefficient tell you about the strength of the relationship?
3) How does Pearson's correlation coefficient differ from covariance?